

Auteur: Milo Rampelbergh
 Studierichting: Informatica
 Oefeningenreeks 2D nr. 6.2

$$f(x) = \frac{x^2 - 5x + 6}{x^3 - 2x^2 - x + 2}$$

$$1. \lim_{x \rightarrow 2} \frac{x^2 - 5x + 6}{x^3 - 2x^2 - x + 2} = \frac{0}{0}$$

$$x^2 - 5x + 6 = 0 \rightarrow x = 3 \text{ of } x = 2$$

$$\begin{array}{c|ccc|c} & 1 & -2 & -1 & 2 \\ 2 & & 2 & 0 & -2 \\ \hline & 1 & 0 & -1 & 0 \end{array}$$

$$= \lim_{x \rightarrow 2} \frac{(x-3)(x-2)}{(x-2)(x^2-1)}$$

$$= \lim_{x \rightarrow 2} \frac{(x-3)}{(x^2-1)} = \frac{-1}{3}$$

$$2. \lim_{x \rightarrow 3} \frac{x^2 - 5x + 6}{x^3 - 2x^2 - x + 2} = \frac{-15 + 6}{27 - 18 - 3 + 2} = \frac{0}{2} = 0$$

$$3. \lim_{x \rightarrow -1} \frac{x^2 - 5x + 6}{x^3 - 2x^2 - x + 2} = \frac{1 + 5 + 6}{-1 - 2 + 1 + 2} = \frac{12}{0} = \infty$$

$$\Rightarrow \lim_{x \rightarrow -1+} \frac{x^2 - 5x + 6}{x^3 - 2x^2 - x + 2} = \lim_{x \rightarrow -1+} \frac{(x-3)(x-2)}{(x-2)(x^2-1)} = \frac{12}{0^+} = +\infty$$

$$\lim_{x \rightarrow -1-} \frac{x^2 - 5x + 6}{x^3 - 2x^2 - x + 2} = \lim_{x \rightarrow -1-} \frac{(x-3)(x-2)}{(x-2)(x^2-1)} = \frac{12}{0^-} = -\infty$$

$$4. \lim_{x \rightarrow \infty} \frac{x^2 - 5x + 6}{x^3 - 2x^2 - x + 2} = \lim_{x \rightarrow \infty} \frac{x^2}{x^3} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

$$5. \lim_{x \rightarrow 1} \frac{x^2 - 5x + 6}{x^3 - 2x^2 - x + 2} = \frac{1 - 5 + 6}{1 - 2 - 1 + 2} = \frac{2}{0} = \infty$$

$$\Rightarrow \lim_{x \rightarrow 1+} \frac{x^2 - 5x + 6}{x^3 - 2x^2 - x + 2} = \lim_{x \rightarrow 1+} \frac{(x-3)(x-2)}{(x-2)(x^2-1)} = \frac{2}{0^+} = -\infty$$

$$\lim_{x \rightarrow 1-} \frac{x^2 - 5x + 6}{x^3 - 2x^2 - x + 2} = \lim_{x \rightarrow 1-} \frac{(x-3)(x-2)}{(x-2)(x^2-1)} = \frac{2}{0^-} = +\infty$$

References